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Compatibility of Spectral and Operator Reductions of a First-Principles Silicon Hamiltonian

Exact rank-two compatibility geometry and a rank-10 silicon Wannier case study

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SECTION I

Introduction

Representations, retained subspaces, and finite-range cost

The time-independent Schrödinger equation

$$\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle$$

Spectral representation

In a finite-dimensional numerical representation,

$$\hat{H} = \sum_n E_n|\psi_n\rangle\langle\psi_n|.$$

The operator is represented by its eigenvalues and eigenvectors.

Real-space representation

$$H(\mathbf{r},\mathbf{r}') = \langle\mathbf{r}|\hat{H}|\mathbf{r}'\rangle,$$

$$\int H(\mathbf{r},\mathbf{r}')\psi_n(\mathbf{r}')\,d\mathbf{r}' = E_n\psi_n(\mathbf{r}).$$

The same operator is represented as a coordinate-space kernel.

The spectrum and the real-space matrix describe the same operator, but they expose different information.

From full reconstruction to a retained subspace

Full and rank-N reconstruction

$$\hat{H}_M = \sum_{n=1}^{M} E_n |\psi_n\rangle\langle\psi_n|,$$

$$P_N = \sum_{n\in\mathcal{W}} |\psi_n\rangle\langle\psi_n|,$$

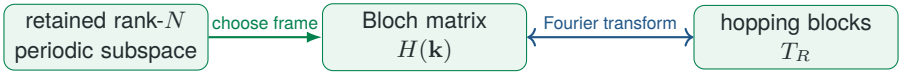
$$\hat{H}_N = P_N \hat{H}_M P_N, \qquad N < M.$$

Rank $N < M$

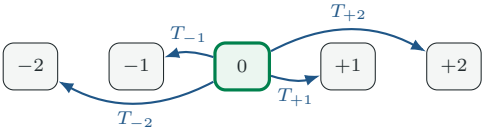
- the DFT–PAW parent can be high dimensional;
- the target may occupy a frontier-band window; and
- a retained rank-N model can make interpolation and downstream analysis more tractable.

Projection remains a separate approximation that must be assessed.

From the retained periodic subspace to hopping blocks



The model: matrix-valued hoppings between cells



Hopping block T_R

A matrix coupling the retained orbital coordinates in cell 0 to those in cell R. Hermiticity requires $T_{-R} = T_R^\dagger$.

Complete periodic model

$$H(\mathbf{k}) = \sum_R T_R e^{i\mathbf{k}\cdot R}.$$

All hopping blocks contribute to the complete Bloch Hamiltonian.

Finite-range reduction

$$H^{(R_c)}(\mathbf{k}) = \sum_{|R|\le R_c} T_R e^{i\mathbf{k}\cdot R}.$$

Increasing R_c retains more real-space information and usually lowers the truncation error.

At a fixed error target, the range cost is the smallest sampled cutoff R_c for which the truncated model passes.

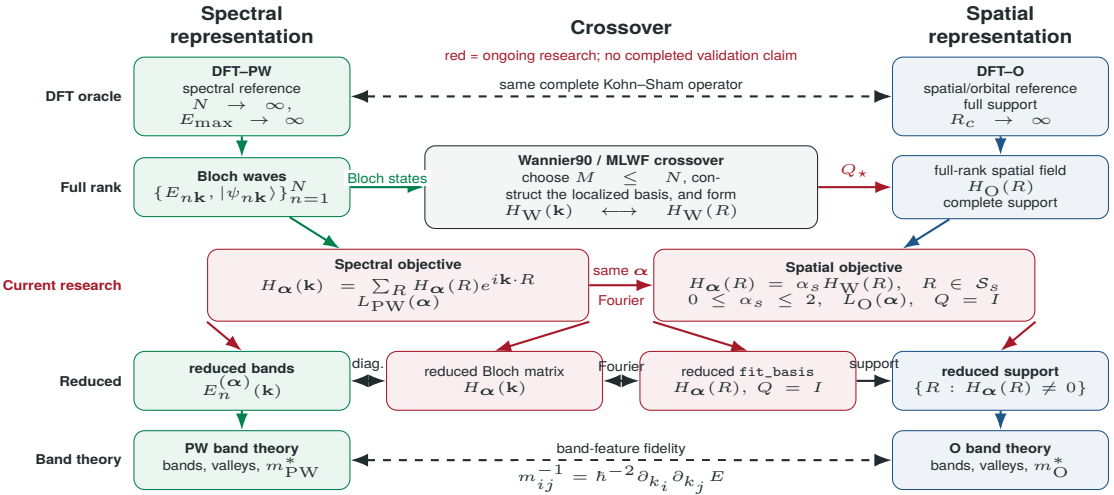
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SECTION II

Scientific Software Implementation

Strong types, formal checks, and operator-guided LLM development

Spectral and spatial reductions in a common fit basis



Research background and implementation capabilities

Education

- BS Mathematics and BS Economics, U.S. Military Academy, West Point;
- MA Mathematics, specialization in Mathematical Finance, Columbia University;
- MSES Mechanical Engineer, U. Pacific;
- PhD Materials Science and Engineering, U. Florida.

Experience and research

- computer consultant: network design, optimization, and deployment;
- quantitative analyst;
- master's research: viscoelastic materials and long-term stress–strain prediction;
- PhD research: DFT, molecular dynamics, potential development, and lattice dynamics.

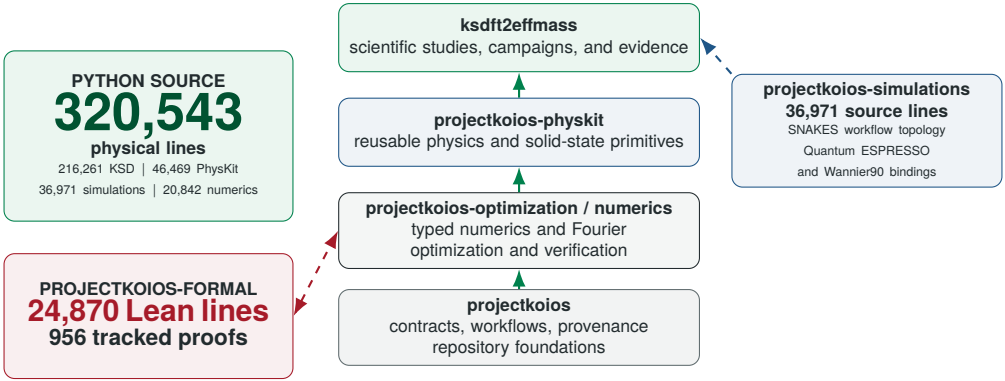
Skills mapped into this program

Mathematics	compatibility geometry, proofs, Lean
Quantitative analysis	optimization, model selection, diagnostics
Computing systems	architecture, tools, deployment, provenance
Mechanics	constitutive behavior and reduced models
Materials science	DFT, Wannier, MD, potentials, phonons

Minimum one-person capability

Domain physics, mathematics, numerics, software engineering, formal/software verification, provenance, and scientific judgment must remain connected. The LLM accelerates implementation; it does not replace expertise, review, or scientific authority.

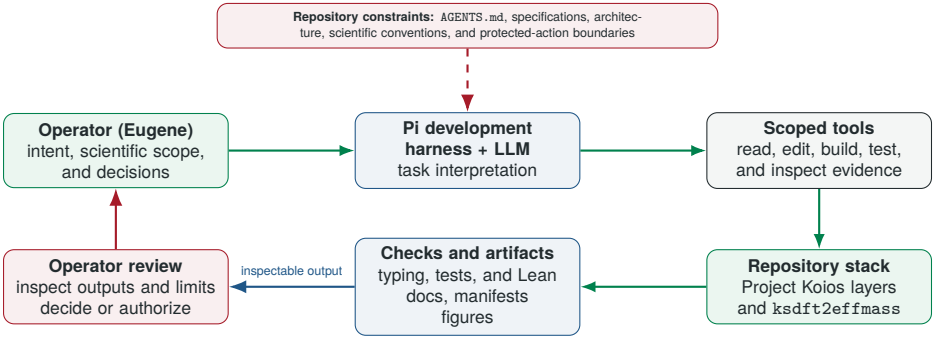
Implementation scale: scientific software stack



At this implementation scale, ad hoc line-by-line inspection is insufficient. A viable one-person program requires layered abstractions, automation, strong typing, tests, formal checks, and human scientific authority.

Counts use selected Python `src/` trees and the `projectkoios-formal` source and theorem-manifest inventories.

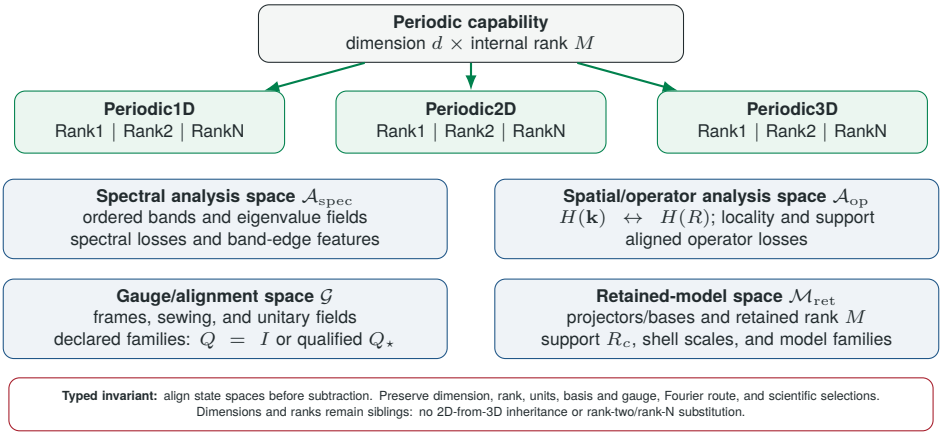
Human-in-the-loop control of LLM development



Human and formal checks constrain nondeterministic output

Human-in-the-loop review, strong typing, tests, and selected Lean formalization make LLM-generated changes inspectable and auditable; they do not make the LLM deterministic. The operator retains scientific and protected-action authority, and passing software checks is not scientific validation.

Strong types encode periodic and analysis spaces



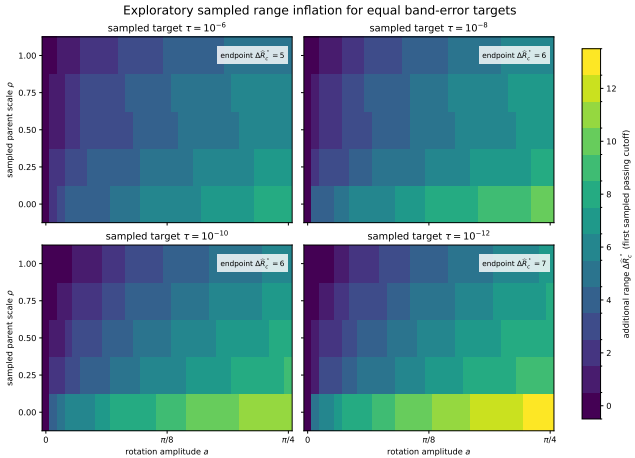
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SECTION III

Rank-Two Periodic Theory

Exact compatibility geometry in periodic dimension d

One band structure, two finite-range costs



Before truncation: identical bands by pointwise unitary similarity.

Keep hoppings with $|R| \leq R_c$ and find the first sampled R_c reaching band error $\leq 10^{-8} E_{\mathrm{ref}}$.

Smallest passing range

Reference	$R_c = 4$
Rotated comparison	$R_c = 10$

Range cost

The rotated comparison needs six additional lattice steps.

How identical bands produce different hopping locality

Pointwise spectral equivalence

$$H'(\mathbf{k}) = U^\dagger(\mathbf{k}) H(\mathbf{k}) U(\mathbf{k}),$$

$$\operatorname{spec} H'(\mathbf{k}) = \operatorname{spec} H(\mathbf{k}).$$

A conventional band plot cannot distinguish the fields.

Fourier representation

$$H'(\mathbf{R}) = \frac{1}{|\mathrm{BZ}|} \int_{\mathrm{BZ}} U^\dagger H U \, e^{-i\mathbf{k}\cdot\mathbf{R}} \, d\mathbf{k}.$$

Momentum dependence in $U(\mathbf{k})$ mixes Fourier modes and redistributes hopping power over $\mathbf{R}$.

Under a gauge-aware identification

The two fields represent the same abstract operator, and the aligned residual vanishes.

Under one fixed identification

Matrix agreement and finite-range cost become additional requirements beyond band agreement.

Spectral equivalence does not determine finite-range locality. Under a fixed orbital identification, matrix agreement and locality become additional constraints.

From two error budgets to one compatibility question

Spectral admissibility

$$\mathcal{A}_s(\tau_s) = \{p : L_{\mathrm{spec}}(p) \le \tau_s\}.$$

Which candidates reproduce the bands within the declared spectral budget?

Operator admissibility

$$\mathcal{A}_o(\tau_o) = \{p : L_{\mathrm{op},*}(p) \le \tau_o\}.$$

Which candidates reproduce the represented operator after the allowed alignment?

Joint match

$$\mathcal{A}_s \cap \mathcal{A}_o \neq \varnothing$$

One candidate passes both error budgets.

Certified separation

$$\mathrm{dist}(\mathcal{A}_s, \mathcal{A}_o) > \rho_{\mathrm{sep}}$$

A verified bound rules out any candidate satisfying both budgets within the declared family.

Unresolved

Neither statement is established.

“No witness found” is not a separation proof.

Periodic-d rank-two proof in arbitrary periodic dimension

$\mathbf{q} \in \mathbb{T}^d$, $\mathbf{R} \in \mathbb{Z}^d$; the periodic dimension d is arbitrary, while the internal fiber rank remains two.

Setup and admitted alignment

$$H(\mathbf{q}) = \bar{E}(\mathbf{q}) I_2 + D(\mathbf{q}), \qquad \operatorname{tr} D = 0,$$

$$A(\mathbf{q}) = U(\mathbf{q})^{\mathsf{T}} H(\mathbf{q}) U(\mathbf{q}), \qquad U(\mathbf{q}) \in SO(2),$$

$$C_{s,\lambda}(\mathbf{q}) = \bar{E} I_2 + s E_{\mathrm{ref}} I_2 + \lambda D.$$

The comparison admits one global $R(\phi) \in SO(2)$, not the generating momentum-dependent field $U(\mathbf{q})$.

Four exact steps

1. $\langle I_2, D \rangle_F = 0$ separates centroid and splitting errors.
2. Conjugation by $R(\phi)$ rotates the real traceless coordinates through 2ϕ; maximizing one weighted correlation gives $\mu_{\max}$.
3. Completing the square gives

$$L^2_{\mathrm{spec}} = s^2 + \kappa(\lambda - 1)^2,$$

$$L^2_{\mathrm{op},*} = s^2 + \kappa(\lambda - \mu_{\max})^2 + \kappa(1 - \mu^2_{\max}).$$

4. Thresholding the two quadratics produces homothetic ellipses; periodic dimension enters only through the averages defining κ and $\mu_{\max}$.

Exact conclusion: $Q = \mathrm{diag}(1, \kappa)$, $c_s = (0, 1)$, $c_o = (0, \mu_{\max})$, and the operator floor is $\kappa(1 - \mu^2_{\max})$. This is periodic-d and internal-rank-two; the locality identity needs an additional flat-parent premise.

Constructing the admissible sets

Inputs: $(\kappa, \mu_{\max}, \tau_s, \tau_o, \mathcal{D})$ with $\mathcal{D} \subseteq \{(s, \lambda) : \lambda \geq 0\}$.

1. Spectral admissible set

$$\mathcal{A}_S = \mathcal{D} \cap \{s^2 + \kappa(\lambda - 1)^2 \leq \tau_s^2\}.$$

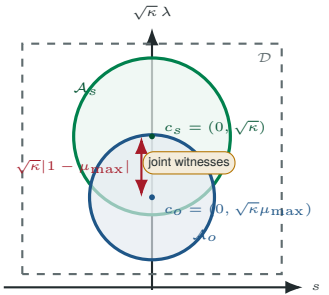
$c_S = (0, 1)$ and $r_S = \tau_S$ in scaled coordinates.

2. Operator floor and set

$$f_o = \kappa(1 - \mu_{\max}^2).$$

If $\tau_o^2 < f_o$, then $\mathcal{A}_o = \varnothing$. Otherwise

$$r_o^2 = \tau_o^2 - f_o,$$
$$\mathcal{A}_o = \mathcal{D} \cap \{s^2 + \kappa(\lambda - \mu_{\max})^2 \leq r_o^2\}.$$



In $(s, \sqrt{\kappa}\lambda)$ coordinates, the ellipses become circles.

Joint match: $\mathcal{A}_S \cap \mathcal{A}_O \neq \varnothing$
contained ellipses: $\sqrt{\kappa}|1 - \mu_{\max}| \leq \tau_S + r_O$

Certified separation: verify
$\mathrm{dist}(\mathcal{A}_S, \mathcal{A}_O) > \rho_{\mathrm{sep}}$

Clipped $\mathcal{D}$: test the constrained sets;
the center-distance condition alone is insufficient

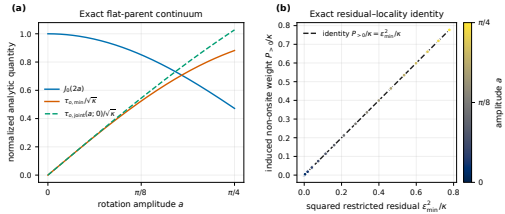
Locality corollary: residual equals non-onsite power

Exact flat-parent result

$$\inf_{s,\lambda\geq 0,\phi} L_{\mathrm{op},\phi}^2 = \kappa(1-\mu_{\max}^2) = \frac{1}{2E_{\mathrm{ref}}^2} \sum_{\mathbf{R}\neq\mathbf{0}} \|B_{\mathbf{R}}\|_F^2$$

Best global $SO(2)$ fit $\rightarrow B_{\mathbf{0}}$; Parseval remainder $\rightarrow$ all $\mathbf{R} \neq \mathbf{0}$ blocks.

$d = 1$ plotted illustration



Periodic-d sine family

$$U_a^{(d)}(\mathbf{q}) = R\left[\frac{a}{d}\sum_{j=1}^{d}\sin(2\pi q_j)\right],$$

$$\mu_{\max}^{(d)}(a) = J_0(2a/d)^d.$$

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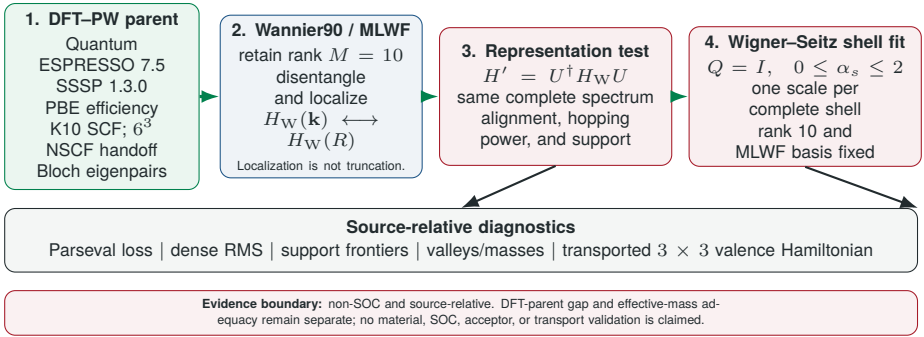
SECTION IV

Rank-10 Silicon

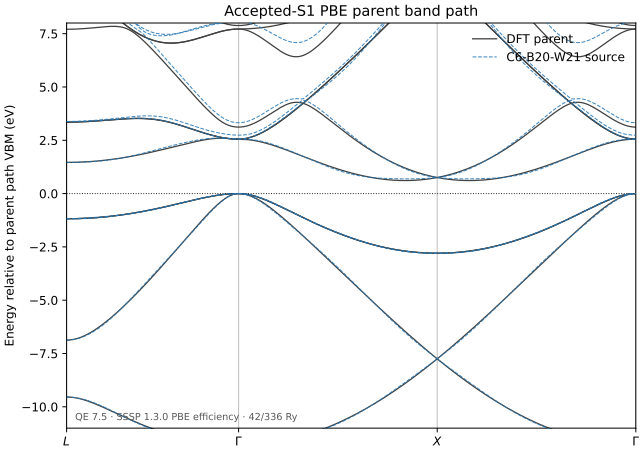
Source-relative Wannier reduction and band-edge diagnostics

Exact finite separation extends to every retained dimension $N \geq 2$;
the two-point construction and its limitations are in backup.

From DFT to Wannier90 to this reduction study



Can the mechanism survive in rank 10?



Source

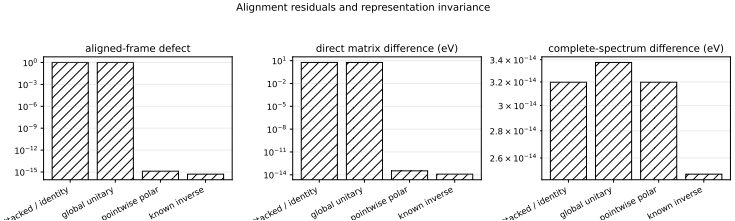
Non-SOC rank-10 $C6$–$B20$–$W21$ silicon Wannier Hamiltonian on a 6^3 construction mesh.

Test

Apply a frozen momentum-dependent Givens circuit and compare alignment, Fourier power, support, interpolation, and band-edge consequences.

Boundary: The source fails separate DFT-parent gap/effective-mass targets.

A unitary change of frame alters locality, not the spectrum



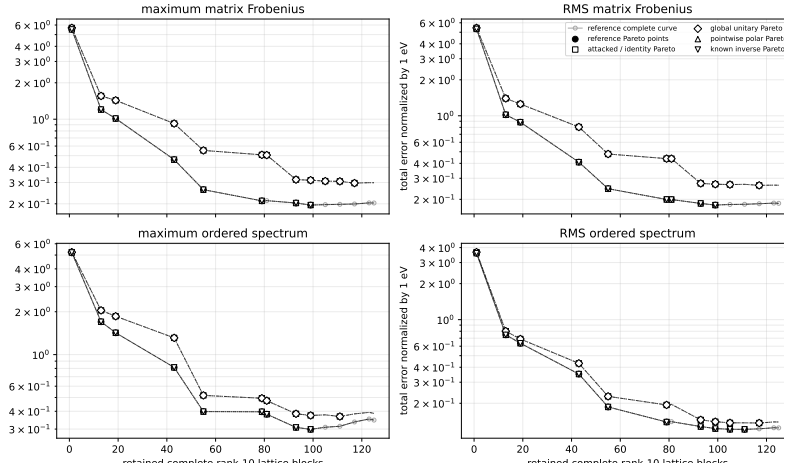
Spectrum check

Maximum sorted-spectrum difference:
3.20×10^{-14} **eV.**

Non-onsite power

$20.7574 \rightarrow 21.5057$ **eV^2/retained state.**

Support–error frontiers depend on representation



Support–error tradeoff

Horizontal position is the number of retained complete rank-10 lattice blocks; vertical position is total error. Lower and farther left is better.

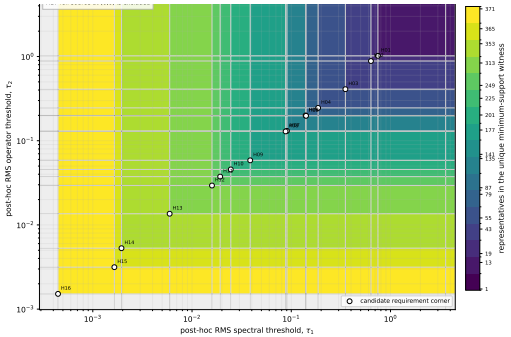
Representation effect

The non-dominated observed supports shift in the matrix and ordered-spectrum channels after the unitary change of frame.

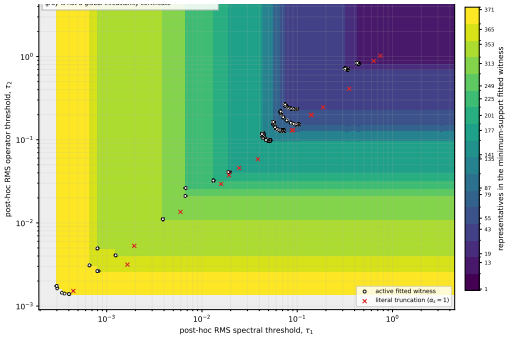
The complete spectrum is unchanged, but the finite-support tradeoff moves.

Boundary. These are discrete finite-inventory frontiers, not certified global optima or infeasibility results.

Wigner–Seitz deletion versus shell fitting



Literal complete Wigner–Seitz shell deletion



Wigner–Seitz shell-scale fitting, $0 \leq \alpha_s \leq 2$

- The bounded multistart search produced 444 converged optimization witnesses; convergence does not by itself establish feasibility.
- Gray regions mark “no witness found”; they do not certify global infeasibility.
- H-shell-16 retains $371/383 = 96.9\%$ of representatives: near-full support, not a compact Hamiltonian.

Fit-sample error understates dense and integrated error

Spectral RMS: 408 points → dense 32^3 mesh

Model	Fit (eV)	Dense (eV)	Increase
H-shell-05	0.07242	0.18196	$2.51\times$
H-shell-10	0.01320	0.02785	$2.11\times$
H-shell-16	0.000404	0.000657	$1.62\times$

Operator RMS: 408 points → Parseval integral

Model	Fit (eV)	Parseval (eV)	Increase
H-shell-05	0.12814	0.26004	$2.03\times$
H-shell-10	0.03229	0.05956	$1.84\times$
H-shell-16	0.001394	0.001747	$1.25\times$

What changes off the fit set?

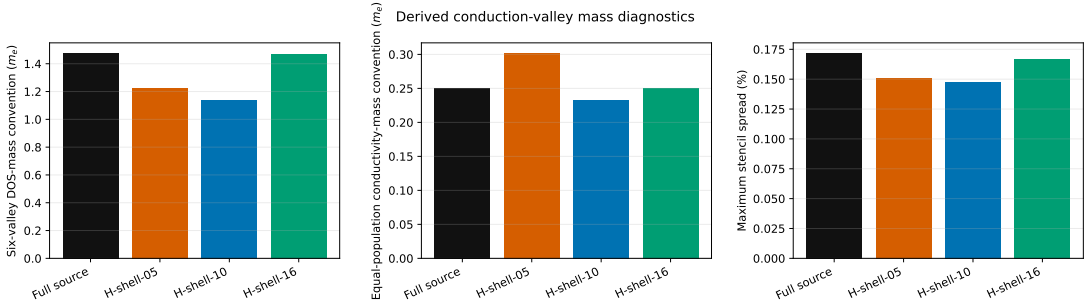
Every reported error increases when evaluated on a denser or integrated domain. The 408 optimization points therefore understate broader source-relative error.

Independent arithmetic check

Directly sampled operator RMS and exact Parseval evaluation agree to 5.56×10^{-17} eV when evaluated under the same convention.

Fit loss, dense interpolation, and band-edge fidelity are distinct tests.

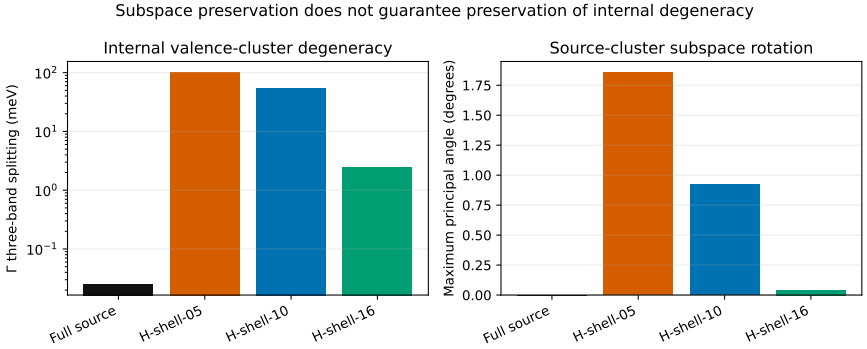
A small spectral RMS can hide a large mass error



H-shell-10: spectral RMS $= 0.02785$ eV, yet the maximum principal electron-mass change is 53.2%.

Context: H-shell-05 changes by 67.7%; H-shell-16 by 2.54%. A good-looking dense-band RMS is not a band-edge certificate.

Valence degeneracy is an internal-fidelity test



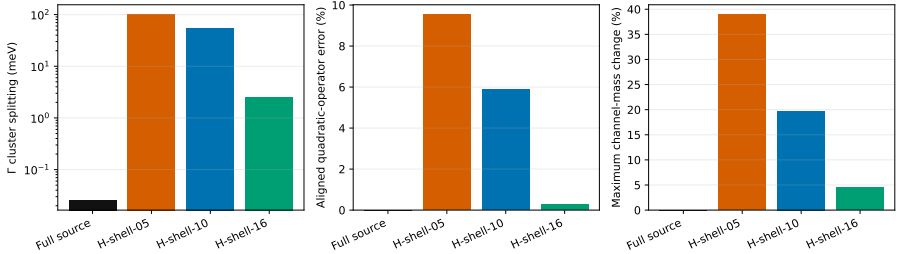
Gamma splitting: source 0.025 meV | H-shell-05 102.2 meV | H-shell-10 54.8 meV | H-shell-16 2.50 meV.

Subspace overlap can remain good while the internal Hamiltonian of the nearly degenerate cluster is substantially distorted.

The correct valence object is matrix-valued

Transport the three-band cluster into one common coefficient space: $H_{\mathrm{eff}}(\mathbf{q}) = H_{\mathrm{eff}}(0) + \sum_i q_i A_i + \sum_{ij} q_i q_j B_{ij} + O(\|\mathbf{q}\|^3)$. For direction $\hat{\mathbf{n}}$, the eigenvalues of $2\sum_{ij} n_i n_j B_{ij}$ are basis-invariant curvature channels.

Non-SOC multiband valence fidelity separates three failure modes



Preserve both the transported subspace and its internal 3×3 quadratic Hamiltonian.

Non-SOC orbital curvature channels—not physical heavy-hole, light-hole, or split-off masses.

The source can fail upstream and still teach us downstream

Deliberate numerical stress test

The DFT-derived rank-10 Wannier operator was subjected to severe complete-shell deletion and shell-scale fitting to expose representation, truncation, and feature-fidelity errors.

Physics scope

The calculations are non-SOC. They do not support physical heavy-hole, light-hole, split-off, acceptor, spin-dependent, or transport claims.

Mesh roles must remain separate

The accepted parent SCF uses a Gamma-centered K10 mesh and the active Wannier source uses a 6^3 construction mesh. The 12^3, 20^3, and 32^3 meshes are post-fit interpolation tests—not a 12^3 parent DFT calculation that was reduced.

Certificate scope

Lean-based certificate checking is not implemented for the silicon admissible sets. Bounded multistart optimization supplies constructive witnesses only; identified valleys and “no witness found” outcomes are not global certificates.

Source adequacy

The full source fails separate DFT-parent gap and most effective-mass targets; no reduction can repair that upstream discrepancy.

Mechanism still testable

The source remains sufficient to test how a frozen unitary change of frame and complete-shell reduction affect alignment, Fourier power, support, and source-relative fidelity.

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SECTION V

Conclusion

Scientific boundaries, take-home messages, and next steps

Take-home messages

1. **Bands do not determine locality.**
2. **Alignment is part of the model definition.**
3. **Fit loss, dense interpolation, and band-edge fidelity are different tests.**
4. **Near-full support is not a compact reduction.**
5. **Source-relative fidelity is not physical validation.**

A reduced Hamiltonian should be accepted against the operator and observables it is intended to reproduce—not against a band plot alone.

What comes next: operator reduction beyond silicon

Wigner–Seitz $\rightarrow$ Slater–Koster

Build a qualified common-space map from the retained Wigner–Seitz hopping representation to an orbital-parametric Slater–Koster basis, then test spectral and operator admissibility together.

Electron–phonon coupling operators

Develop an EPL-like reduction workflow for coupling operators with aligned electronic and phonon state spaces; keep gauge, interpolation, truncation, and observable fidelity as separate error channels.

Doping beyond effective mass

Represent dopant potentials and host–defect coupling as reduced operators, retaining screening, symmetry, localization, finite-domain effects, and provenance rather than only effective-mass tensors.

Conjugated donor–acceptor polymers

Start from the DFT–O orbital representation, construct aligned local operators for donor–acceptor systems, and test spectral/operator compatibility before finite-range reduction.



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Thank you.

Questions?

Band equivalence is not locality.

Backup: evidence status

Layer	Strongest retained statement	Boundary
Analytic rank two	Exact loss geometry and residual–tail identity	Restricted theorem assumptions
Controlled rank two	Sampled range inflation from 4 to 10	Declared unitary field, mesh, target, and cutoff inventory
Rank-10 silicon	Spectrum invariant; locality and finite-support behavior change	Producer checked; not independently reconstructed
Convention audit	Six-valley formulas and exact size-six mode inventory agree	No fields, valleys, Hessians, or removed coefficients reconstructed
DFT context	Accepted-S1 path and revealed 408-point adequacy audit	Context/development evidence, not material validation

Backup: rank-two assumptions

The residual–non-onsite-power identity assumes:

- a flat, traceless, real-symmetric 2×2 parent;
- a generating unitary field in the same real-rotation orbit;
- one global real-rotation alignment and a nonnegative scale;
- normalized continuum measure or uniform finite weights;
- matched Fourier normalization.

The invariant correlation term is nonnegative, so the reported scale optimum is not clipped. For a different orbit with negative maximal correlation, minimization over $\lambda \geq 0$ would stop at $\lambda = 0$.

Backup: controlled numerical protocol

- finite-range four-state parent; retained lower rank-two projector;
- parallel transport with periodic sewing and overlap-rank gates;
- centered construction meshes up to $N = 512$;
- shifted, disjoint evaluation mesh with $N_{\mathrm{eval}} = 1025$;
- ordinary-lattice cutoffs $R_c = 0, \ldots, 16$;
- first-passing decision uses maximum ordered-band error;
- interpolation, truncation, and total finite-model errors retained separately.

Backup: size-six Fourier conversion

Cyclic inventory:

$$\{-3,-2,-1,0,1,2\}^3, \qquad 216 \text{ records.}$$

Remove every record touching $-3 = -N/2$:

$$91 \text{ removed}, \qquad 125 \text{ retained.}$$

Retained modes map identically to $\{-2,-1,0,1,2\}^3$.

Independent audit

FCC metric labels agree; maximum radius discrepancy is 3.55×10^{-15} Å.

Unavailable

The 216 cyclic matrices and removed-mode coefficient power were not retained and were not reconstructed.

Backup: six-valley mass conventions

For valley mass tensor M_v:

$$m_{\mathrm{DOS},1v} = \det(M_v)^{1/3}, \qquad m_{\mathrm{cond},1v} = \frac{3}{\mathrm{tr}(M_v^{-1})},$$

$$m_{\mathrm{DOS},6v} = \left[\sum_v \sqrt{\det M_v}\right]^{2/3}, \qquad m_{\mathrm{cond},6v} = \frac{3}{\mathrm{tr}[(1/6)\sum_v M_v^{-1}]}.$$

Independent arithmetic

Maximum difference from separate NumPy calculations: 6.22×10^{-15}.

The conductivity convention averages inverse-mass tensors, not scalar conductivity masses.

Backup: Slater–Koster is not the active fit

Active reduction family

The reported fit rescales complete native Wigner–Seitz shells,

$$0 \leq \alpha_s \leq 2,$$

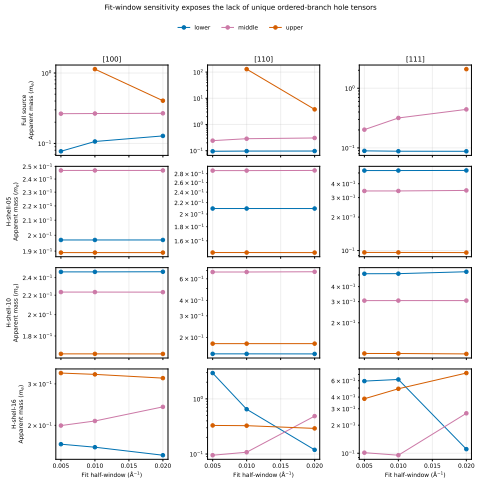
while preserving the rank-10 Wannier basis and all matrices within each shell.

Separate Slater–Koster path

A common-space gate for a joint orbital-aligned Slater–Koster fit was rank deficient (rank 9) at Gamma and one additional point, so no joint fit was run.

This does not establish a Slater–Koster impossibility theorem; it only excludes that unqualified route from the active results [1, 2].

Backup: ordered valence branches are window sensitive



Missing curves correspond to positive fitted curvature, for which no apparent hole mass is reported.

Backup: reproducibility boundary

Retained calculation packages contain requests, source identities, producer scripts, machine-readable results, figures, manifests, and execution records.

Current limitation

No public repository URL or DOI yet contains the complete authenticated source data and independent reconstruction environment. Local paths and hashes are audit anchors, not a public reproduction package.

No new QE, Wannier90, Hamiltonian construction, optimization, valley search, or Hessian calculation was performed to prepare these slides.

Backup: exact separation for every retained dimension

For $N \geq 2$, let $A_N = \mathrm{diag}(\varepsilon_0 < \cdots < \varepsilon_{N-1})$ on $K = \mathbb{Z}/2\mathbb{Z}$, with G_{01} swapping the first two coordinates.

$$U(0) = I, \quad U(1) = G_{01}, \quad B(k) = U(k)^\dagger A_N U(k).$$

Pointwise spectrum

$$\mathrm{spec}_{\mathbb{C}}\, B(k) = \mathrm{spec}_{\mathbb{C}}\, A_N.$$

Truncation does not commute

$$T_0[U^\dagger H_A U] \neq U^\dagger T_0[H_A] U.$$

Positive omitted power

$$P_{\mathrm{omit}}(\widehat{B}; \{0\}) > 0.$$

Generic exact result

If a finite-dimensional operator satisfies $A \neq G^\dagger A G$, the same two-point construction gives all three conclusions with one common witness.

Concrete rank-N witness

The analytic parent contains all N strictly ordered retained energies. This is retained-space dimension N, not algebraic matrix rank or a rank-two embedding.

Exact scope and current status

N is retained-space dimension, not algebraic matrix rank. This is a minimal $\mathbb{Z}/2\mathbb{Z}$ witness with spectrum equality as complex sets; it proves no realistic-mesh, runtime-FFT, continuum-locality, or material statement. Targeted Lean compilation and the no-admission scan pass in a development worktree; promotion and a frozen audit remain pending. IDs: BG-FOURIER-GAUGE-THM-013/014.

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